

DevOps Lab Programs

Ubuntu OS — Step-by-Step Procedures

Program 1: Introduction to Jenkins — Installation & Setup

Step 1: Check Java Installation

```
java --version
```

Step 2: Download Jenkins

1. Open browser → Google
2. Search: download Jenkins in ubuntu
3. Click on link "download and deployment"
4. Scroll down → get two lists
5. Download Jenkins 2.375.1 → Select → Click Ubuntu → Enter
6. Commands are displayed — copy & paste to terminal one by one

Step 3: Install Dependencies

```
sudo apt update
sudo apt install openjdk-17-jdk
sudo apt install eclipse
sudo apt install maven -y
mvn version
```

For Gradle: Go to browser → search steps to install Gradle on GreenCloud → copy & paste to terminal.

```
gradle -v
```

Step 4: Install Jenkins via Terminal

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc https://...
echo "deb [signed-by=...] ..." | sudo tee
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt-get update
sudo apt-get install jenkins
```

Step 5: Enable & Start Jenkins Service

```
sudo systemctl enable jenkins
sudo systemctl start jenkins
```

Step 6: Check Jenkins Service Status

```
sudo systemctl status jenkins
```

Step 7: Setup Post-Installation (Browser)

7. Open browser → type in URL:

```
localhost:8080
```
8. It opens a new window → copy the path shown and paste in terminal to get password:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```
9. Copy the displayed password.

10. Paste password in Jenkins window (browser) → Click Continue

11. Customize Jenkins page appears:

- 1. Install Suggested Plugins
- 2. Select Plugins to Install

12. Click option 1 → Install Suggested Plugins → Wait for installation

13. Sign Up page opens automatically after all plugins install:

First Admin User:

→ **Username:** tadmin

→ **Password:** devulca@0

→ **Confirm Password:** devulca@0

→ **Full Name:** Sutla

→ **Email Address:** sutreme123@gmail.com

14. Press Save & Continue → Jenkins is ready!

Login via: localhost:8080 → Username: tadmin → Password: devulca@0

Program 2: Jenkins — Adding SSH Key Credentials

In Jenkins (localhost:8080) → Login

First, generate SSH Key and create credentials.

Generate SSH Key (Terminal)

| `ssh-keygen`

Copy the SSH public key.

Add SSH Credentials in Jenkins

15. Dashboard
16. Manage Jenkins
17. Select the domain (Global)
18. Add new credentials
19. Click on Add Credentials:
 - **Kind:** SSH Username with Private Key
 - **Username:** Enter git
 - **Scope:** Global
20. Click Private Key → Enter Direct Key → Paste the SSH code

Program 3: Jenkins CI Pipeline with Maven Integration

Step 1: Create Maven Project Directory (Terminal)

```
mkdir abbu  
cd abbu
```

Step 2: Generate Maven Project

```
mvn archetype:generate -DgroupId=com.example -DartifactId=my-app -  
DarchetypeArtifactId=maven-archetype-quickstart  
ls  
cd my-app  
mvn clean install
```

Step 3: Initialize Git Repository

```
git init  
git add .  
git commit -m "pom Example"  
git branch -m main  
git remote add origin <url>  
git push -u origin main
```

Program 4: Maven Project in Eclipse IDE

21. Open Eclipse
22. In Eclipse: Go to File → New → Maven Project
23. Click Next → open new window then select:
 - **GroupId:** org.apache.maven.archetype
 - **ArtifactId:** maven-archetype-quickstart
 - **Version:** 1.1
24. Wait for a few minutes → Click Next
25. In new window, type the following:
 - **GroupId:** com.program2.maven
 - **ArtifactId:** program2-example-jar
 - **Version:** 1.1 (keep snapshot as it is)
 - **Package:** com.program2.maven.program2
26. Check: Run Archetype → Click Finish
27. Configuration confirmation → Y → Build Success
28. Left top of window → program2-example-jar → double click on pom.xml
29. Program appears on screen → Go to Run → Run as → Maven Build → display window
30. Select goal as package → Click Run → Build Success
31. Eclipse left top corner → src/main/java → com.program2.maven.program2 → double click App.java
32. Click Run → Hello World

Program 5: Gradle Project Setup

Step 1 — Create and Initialize Gradle Project

```
mkdir username  
cd username  
gradle init
```

During setup, select:

- Type: 2 (Application)
- Language: 3 (Java)
- Generate using new API [yes/no]: no
- Test framework: 1 (JUnit 4)
 - **Project name:** gradletest
 - **Source package:** gradletest

Step 2 — Build the Project

```
gradle build
```

Expected output: BUILD SUCCESSFUL

Step 3 — Run in Eclipse

Open Eclipse → Ctrl+Alt+Shift+T → Terminal

```
gradle init
```

- 1. Kotlin → Select 2 (Groovy)
- 2. Groovy

Type yes → validating

Generate build using new API [Yes/No] → Yes → Build Successful

```
gradle build
```

Build Successful

Program 6: User Management & Ansible Setup

Step 7: Create New User

```
sudo adduser username
```

Enter new password → Retype password → Enter

```
sudo usermod -aG username
```

```
sudo su username
```

```
ssh-keygen
```

Press Enter

```
hostname -I
```

Get remote host IP (e.g., 10.100.50.90)

```
ssh-copy-id username@<paste-remote-host-ip>
```

Type new password

If SSH is Not Installed

```
sudo apt install openssh-server
```

```
sudo apt install update
```

```
sudo apt install ansible -y
```

```
ansible --version
```

Ansible Configuration

```
sudo mkdir -p /etc/ansible
```

```
sudo nano /etc/ansible/hosts
```

Add the following line:

```
localhost ansible_connection=local
```

Save file & exit

```
ansible-inventory --list -y
```

```
sudo ansible all -m ping
```

Program 7: Ansible Plugin in Jenkins Pipeline

Step 8: Install Ansible Plugin in Jenkins

33. Open Terminal:

```
sudo apt install ansible -y  
pwd
```

Copy the path.

34. Login to Jenkins (localhost:8080):

→ **Username:** tadmin

→ **Password:** siel@0

35. Navigate: Dashboard → Manage Jenkins → Plugins → Available Plugins

36. Search: Ansible → Install Ansible Tower

37. Navigate: Dashboard → Manage Jenkins → Tool → scroll down → Get Ansible section from → Click on it:

→ **name:** ansible

→ **path:** paste the copied path

38. Save → Apply

39. Dashboard → New Item → Enter project name → (Pipeline) → Select Pipeline → Press OK

40. Scroll down → Go to GPT → Ask for simple pipeline program → Copy that program

41. Paste it → Save → Apply → Run

Program 8: Maven Project with Gradle Build Tool

Step 1: Install Dependencies

```
sudo apt update
sudo apt install openjdk-17-jdk
sudo apt install eclipse
sudo apt install maven -y
mvn version
```

Step 2: Install Gradle

Go to browser → search steps to install Gradle on GreenCloud → copy & paste to terminal.

```
gradle -v
```

— End of DevOps Lab Programs —